

Side channel attacks

Learning objectives

- **Describe how attackers exploit physical or indirect information**
- **Explore types like timing, power, and electromagnetic attacks**
- **Examine how to defend against side channel techniques**

What are side channel attacks?

- **Exploit physical signals or behaviours during computation**
- **Do not rely on code flaws or software bugs**
- **Examples: power use, timing, sound, electromagnetic leaks**

Common side channels



Time

Timing analysis uses variations in processing time to infer secret values.



Power

Power analysis looks at changes in energy usage during computation.



Electromagnetic
radiation

Electromagnetic attacks capture data from emitted signals.



Sound

Acoustic analysis uses sound patterns, such as keyboard clicks, to deduce information.

How side channel attacks work

- **Often require close physical proximity**
- **Capture data during sensitive operations like decryption**
- **Use statistical analysis and modelling to recover keys or passwords**

Timing attacks

- **Measure time taken for operations like password checks**
- **Reveal characters based on validation timing**
- **Used against RSA and AES algorithms**

Power analysis attacks

- **Simple: detect patterns in power traces**
- **Differential: compare traces to extract secrets**
- **Affects embedded systems and smartcards**

Electromagnetic and acoustic attacks

- **Electromagnetic (EM) attacks use probes to capture radiation from chips**
- **Acoustic attacks analyse fan noise or keystrokes**
- **Some attacks use phone microphones to gather data**

Real-world cases

- **Smartcard keys extracted via power monitoring**
- **RSA keys captured through walls using EM leaks**
- **Acoustic side channels cracked laptop encryption**

Defences and countermeasures

- **Use constant-time complexity algorithms**
- **Add power and timing noise**
- **Shield hardware to prevent signal leaks**
- **Use secure enclaves for isolated processing**

Summary

- **Side channel attacks use physical observations to break security**
- **They compromise secure systems by measuring power, time, sound or radiation**
- **Countermeasures include hardware protection and code techniques**

Next

- **Explore blockchain and its use in cybersecurity**